A paleomagnetic test for northward tectonic transport of the eastern Selwyn Basin, northern Canadian Cordillera: data from the mid-Cretaceous Ragged Pluton

D.T.A. Symons, K. Kawasaki, C.J.R. Hart, and M.J. Harris

Abstract: Ragged Pluton is a member of the Tungsten Plutonic Suite in the eastern Selwyn Basin. This circular massive 97 Ma granitic pluton is ~5 km in diameter. It intrudes deformed and metamorphosed clastic strata of the Neoproterozoic to Lower Cambrian Vampire Formation. Thermal and alternating field step demagnetization data for 216 specimens from 21 plutonic sites isolated a stable characteristic remanent magnetization (ChRM) direction at 20 sites of declination D = 325.9°, inclination I = 80.4° (α_{95} = 2.7°, k = 142). Thermal demagnetization and saturation remanence analysis show that the ChRM is carried by both single-domain pyrrhotite and magnetite. Ragged Pluton's paleopole at 73.9°N latitude, 191.4°E longitude (dp = 5.0°, dm = 5.2°) is concordant with its co-eval 97 Ma North American reference paleopole, showing at 95% confidence that the pluton has not been significantly rotated or translated relative to North America. This paleomagnetic result favours tectonic models in which the eastern Selwyn Basin is autochthonous, or nearly so, since the mid Cretaceous.

Résumé : Le pluton Ragged est un membre de la suite plutonique Tungsten dans le bassin Selwyn oriental. Ce pluton granitique, circulaire et massif, datant de 97 Ma, a un diamètre ~5 km. Il pénètre les strates clastiques déformées et métamorphosées de la Formation de Vampire (Néoprotérozoïque à Cambrien inférieur). Les données de la désaimantation par les méthodes de champ alternatif progressif et de chauffage par paliers successifs de 216 spécimens provenant de 21 sites plutoniques a permis d'isoler, à 20 sites, une direction d'aimantation rémanente caractéristique (« ChRM ») stable de déclination D = 325,9° et d'inclination I = 80,4° (α_{95} = 2,7°, k = 142). La démagnétisation thermique et l'analyse de la saturation rémanente montrent que la ChRM est portée à la fois par la pyrrhotite et la magnétique à domaines monocristallins. Le paléopôle du pluton Ragged, situé à une latitude de 73,9°N et une longitude de 191,4°E (dp = 5,0°, dm = 5,2°), concorde avec le paléopôle contemporain de référence pour l'Amérique du Nord il y a 97 Ma, montrant donc, avec un degré de confiance à 95 %, que le pluton n'a pas subi de rotation ni de translation importantes par rapport à l'Amérique du Nord. Ce résultat paléomagnétique appuie les modèles tectoniques selon lesquels le bassin Selwyn est autochtone, ou presque, depuis le milieu du Crétacé.

[Traduit par la Rédaction]

Introduction

The inboard portions of the northern Cordillera consist of the carbonate-dominated, Mesoproterozoic to Paleozoic Mackenzie Platform and the clastic-dominated Neoproterozoic to Paleozoic Selwyn Basin that were variably deformed in the Mesozoic into the Mackenzie and Selwyn fold and thrust belts, respectively. Much of the Selwyn Basin was intruded later by suites of Cretaceous plutons with related metamorphic activity. Despite their apparently stable position as miogeoclinal cover on the North American cratonic edge, previous paleomagnetic studies have led to four distinct tectonic scenarios being suggested for the Selwyn Basin since the mid-Cretaceous (Fig. 1).

From the analysis of >40 reliable Mesozoic and Cenozoic paleopoles from the Intermontane Belt (IMB) and Yukon–Tanana (Y–T) terranes, Symons et al. (2005) calculated that the northern IMB terranes were rotated clockwise by 35° ± 14° between 102 ± 14 and 54 Ma and by a further 16° ± 6° since 54 Ma. From this rotation, they estimated an ENE–WSW shortening of 305 ± 125 km across the Selwyn Basin and Mackenzie Mountains between the IMB and ancestral North America in southeastern Yukon and southwestern Northwest Territories. Two alternatives could accommodate this shortening. One option assumes thick-skin tectonics caused the Selwyn Basin to be deformed as it was driven east-northeastward, perhaps contributing to the arcuate trend

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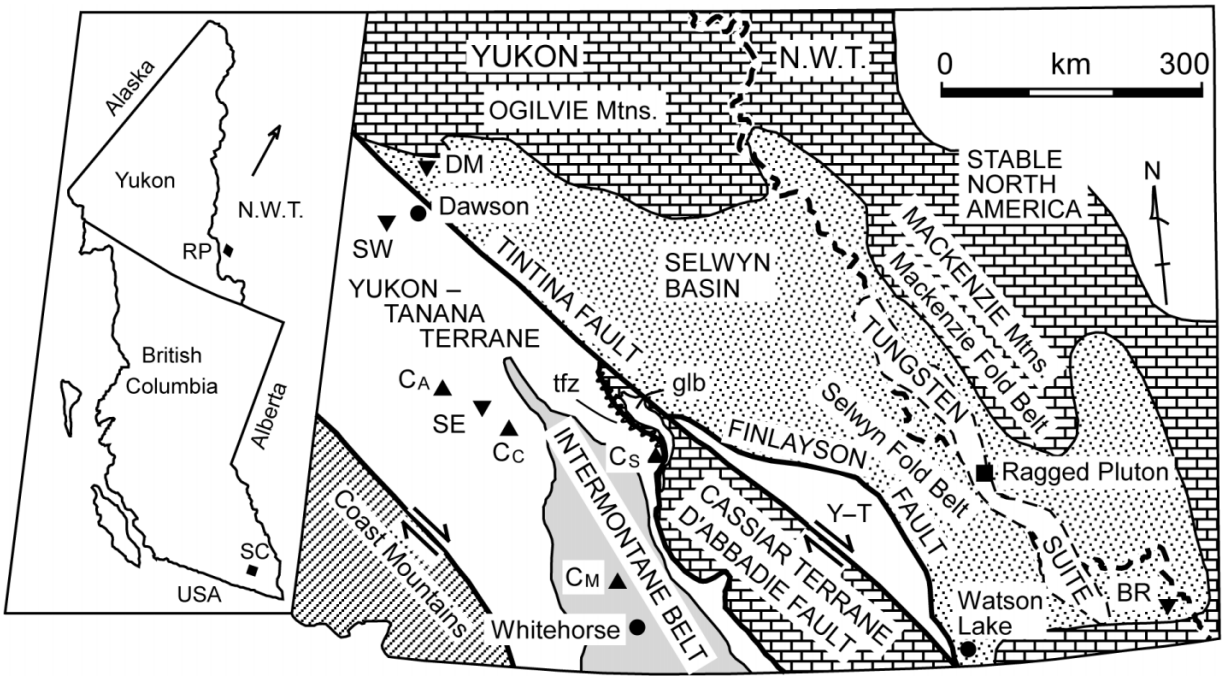
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D.T.A. Symons[1] and K. Kawasaki. Department of Earth and Environmental Sciences, University of Windsor, Windsor, ON, N9B 3P4, Canada.
C. Hart.[2] YT Geological Survey, Government of YT, Box 2703 [K-10], Whitehorse, YT Y1A2C6, Canada.
M. Harris. Department of Geology and Environmental Science, James Madison University, MSC 7703, Harrisonburg, VA 22807, USA.

[1]Corresponding author (e-mail: dsymons@uwindsor.ca).
[2]Present address: School of Earth and Geographical Sciences, University of Western Australia, 35 Stirling Highway, Crawley, WA6009, Australia.

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Fig. 1. Location map. BR, Beaver River; C_A, C_C, C_M, C_S, Carmacks volcanics at Apex Mountain, Carmacks, Miners Range, and Solitary Mountain, respectively; DM, Deadman Pluton; glb, Glenlyon Batholith; RP, Ragged Pluton (inset map); SC, Skelly Creek Batholith (inset map); SE, Seymour Stock; SW, Swede Dome Stock; tfz, Tummel fault zone; Y–T, Yukon–Tanana terrane.



of the Mackenzie Mountains. Alternatively, assuming thin-skin tectonics, the IMB terranes could have slid over the Selwyn Basin and since been eroded without significantly deforming its strata. Both options should give paleomagnetic paleopoles showing the Selwyn Basin to be autochthonous or para-autochthonous, except that the thick-skin option should also show a clockwise rotation.

Based on paleomagnetic results from the ~70 Ma Carmacks Group volcanics, Enkin et al. (2006) suggested that the Selwyn Basin strata originated ~1950 ± 600 km further south relative to ancestral North America. The Carmacks volcanics appear to rest on rocks of the IMB and Y–T terranes (Fig. 1), leading Enkin et al. (2006) to infer that the Selwyn Basin's strata were allochthonous to North America and part of the far-traveled Baja British Columbia microplate (Irving et al. 1996). If true, then mid-Cretaceous rocks in the basin should have significantly shallower characteristic remanent magnetic (ChRM) inclinations than those of North America.

In southeastern British Columbia, Irving and Archibald (1990) analysed paleomagnetic data from the massive ~97 Ma Skelly Creek Batholith in a similar geologic setting on the North American margin (Fig. 1) (Gabrielse et al. 1991). They concluded that the batholith is close to its present location relative to ancestral North America, thereby supporting the traditional geologic viewpoint that the batholith is an autochthonous, or at least para-autochthonous, part of North America since ~97 Ma (Gabrielse 1967; Gordey and Anderson 1993).

Here we report paleomagnetic results from the ~97 Ma Ragged Pluton, the first such results from the easternmost Selwyn Basin (Fig. 1). Our objective was to provide an initial test of the proposed tectonic models for the basin.

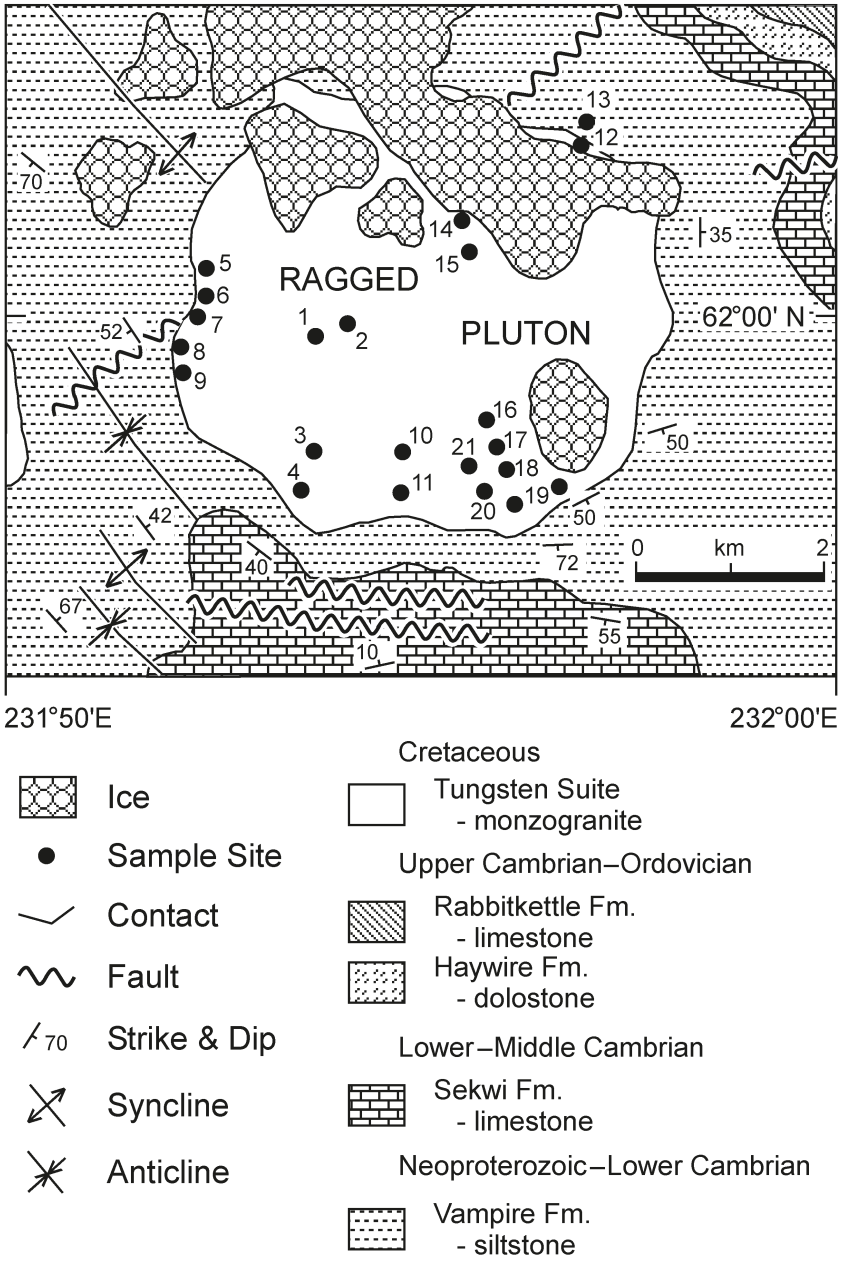
Geology

The stratigraphic sequence in southeastern Yukon and southwestern Northwest Territories comprises a westward-thickening wedge of mid-Proterozoic to mid-Jurassic miogeoclinal sedimentary rocks that are similar to those deposited elsewhere along the western Cordilleran margin of North America (Gordey and Anderson 1993). Although characterized as a passive margin assemblage, this sequence records a number of uplift, erosional, and rifting events. In the mid-Jurassic, an island-arc terrane from the southwest collided against the miogeocline that caused imbrication and folding of the sedimentary succession through the Early Cretaceous, forming two distinct fold belts (Gordey and Anderson 1993). On the northeastern side, the competent carbonate-dominated shallow-water platformal strata deformed to create the Mackenzie Fold Belt (Fig. 1). On the southwestern side, in contrast, the incompetent shale-dominated deeper water basinal sediments were intensely folded with slaty cleavage to form the Selwyn Fold Belt. In the mid-Cretaceous, five plutonic suites were emplaced into the Selwyn Fold Belt, i.e., 109–95 Ma Anvil, 96–93 Ma Mayo, and 92–90 Ma Tombstone suites in the northern and western Selwyn Basin and the 106–96 Ma Hyland and 98–94 Ma Tungsten suites in the southern and eastern basin (Hart et al. 2004). The intrusion of the plutonic suites has been attributed to crustal thickening from deformation and thrust sheet stacking in the basin (Gordey and Anderson 1993; Mair et al. 2006).

Ragged Pluton, a member of the Tungsten Plutonic Suite and remote from older Hyland plutons, is composed of massive 98 to 94 Ma aluminous granitic intrusions that outcrop in an ~50 km wide belt for ~300 km along the Yukon – Northwest Territories border in the easternmost

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Fig. 2. Geology of Ragged Pluton with sampling sites (numbered black dots).



Selwyn Basin (Fig. 1; Hart et al. 2004). The age of the pluton is ~97 Ma based on U–Pb ages from adjacent plutons (Heffernan 2004) and an unpublished sensitive high-resolution ion microprobe (SHRIMP) U–Pb age (C.J.R. Hart, personal communication, 2007). The Tungsten suite plutons are predominantly reduced, peraluminous muscovite–biotite monzogranites (Gordey and Anderson 1993; Hart et al. 2005) with low Fe_2O_3/Fe ratios of ~0.1–0.3, high ilmenite content, moderate to low magnetic susceptibilities, and low aeromagnetic expression (Hart et al. 2004). Typically, these plutons have homogeneous compositions, massive undeformed textures, steep sides, and flat tops, and were likely emplaced at depths of ~5 to 7 km.

Ragged Pluton is circular in plan with a diameter of ~5 km (Fig. 2). It intrudes the Neoproterozoic to Lower Cambrian Vampire Formation, which undergoes a lateral facies transition to the Hyland Group (Gordey and Anderson 1993; Gordey and Makepeace 2003). The Vampire Formation is composed of brown to greenish-grey siltstones with minor fine-grained quartz sandstones and shales that have been metamorphosed locally to orthoquartzite, phyllite, slate, and argillite. Near Ragged Pluton, the Vampire Formation is conformably overlain by carbonates of the Lower to Middle Cambrian Sekwi Formation and by Upper Cambrian to Lower Ordovician limestone formations, which are conformably overlain in turn by undifferentiated strata of the Paleozoic Road River Group.

Structurally, Ragged Pluton was emplaced into the crest of an open anticline with contacts that crosscut bedding. Although there are numerous small-offset north- and northeast-trending normal faults in the region, the only potentially significant post-mid-Cretaceous displacements are west-side-

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up offsets along the March Fault, located ~20 km southwest of Ragged Pluton (Hart and Lewis 2006).

Sampling and measurement

Cores, oriented by sun or magnetic compass, were collected from 20 sites distributed throughout Ragged Pluton and from one site (site 13) in its contact aureole (Fig. 2) using helicopter access from the Cantung mine site, which is about 10 km due west of the pluton (Fig. 1). From these 2.54 cm diameter cores, 216 standard paleomagnetic specimens were sliced. After six months of storage in a magnetically shielded room to allow their viscous remanent magnetization (VRM) components to decay, the natural remanent magnetization (NRM) components of the specimens were measured using an automated vertical-axis 2G Enterprises DC-SQUID magnetometer with a specimen sensitivity of $\sim 2 \times 10^{-6}$ A/m. Initially one specimen per site was alternating field (AF) demagnetized in 13 steps up to 120 mT using a Sapphire Instruments SI-4 demagnetizer and a second specimen per site was thermally (Th) demagnetized in 14 steps to 610 °C using a Magnetic Measurements MM-80 demagnetizer. Saturation isothermal remanent magnetization (SIRM) testing was done on 12 representative specimens by AF demagnetizing them first at 130 mT, then pulse magnetizing in 12 steps up to 900 mT using a Sapphire Instruments SI-6 DC pulse magnetizer, and then AF demagnetizing them in six steps up to 120 mT.

Results

The specimens from the 20 sites in Ragged Pluton had a median NRM intensity of 2.20×10^{-3} A/m (lower and upper quartiles 1.09×10^{-3} and 5.48×10^{-3} A/m). In contrast, site 13 specimens from the contact aureole gave a median intensity of 2.03×10^{-2} A/m, or ~10 times more intense, consistent with the pluton exhibiting an aeromagnetic low anomaly.

Thermal step demagnetization of the 21 Th pilot specimens mostly showed a northerly and steeply downward ChRM direction up to ~560–585 °C. However, the ChRM intensity in these specimens dropped sharply first between ~260 and 340 °C and then again between ~500 and ~585 °C, leaving <2% of the original NRM intensity (Fig. 3), indicating the demagnetization first of either pyrrhotite probably or titanomaghemite possibly around 300 ± 40 °C and then the demagnetization of magnetite above 500 °C. The concordant ChRM directions of the two minerals suggest that their ChRMs are coeval. Overall, pyrrhotite contributes from ~50% to ~90% of the ChRM intensity, and magnetite contributes the remainder.

AF step demagnetization of the 21 AF pilot specimens up to 120 mT yielded the same ChRM directions as found for the Th pilot specimens (Fig. 4), and so all of the remaining undemagnetized specimens were AF demagnetized in 13 steps to 125 mT.

The saturation remanence results (Fig. 5) reflect mixed magnetic mineralogy. The rapid acquisition of SIRM to >0.95 by ~200 mT (Fig. 5*a*) of the site 1 specimen and its rapid intensity decay on AF demagnetization (Fig. 5*b*) is typical of multidomain to pseudosingle domain magnetite or multidomain pyrrhotite. The moderately rapid acquisition and decay curves exhibited by the specimens from sites 12 and 18 are indicative of pseudosingle to single domain magnetite or pseudosingle domain pyrrhotite. Except for the site 10 specimen with a slow rate of SIRM acquisition that is typical of hematite, the SIRM curves for the remaining specimens suggest dominantly single domain pyrrhotite with subsidiary single domain magnetite.

The specimen ChRM directions were determined using vector component analysis (Kirschvink 1980) and remagnetization circle analysis (Bailey and Halls 1984), and the site mean ChRM directions were calculated following Fisher (1953) (Table 1). At site 3, no ChRM endpoint directions were found. The remaining "omitted" specimens listed in Table 1 gave anomalous endpoint directions, anomalous remagnetization circles, or incoherent data. The host rock (site 13) in the pluton's contact aureole gives a ChRM direction that is statistically the same at 95% confidence as the adjacent plutonic site (site 12), and so it has been added to the pluton's population of site mean directions (Fig. 6). Thus Ragged Pluton's mean ChRM direction is $D = 325.9°$, $I = 80.4°$ ($N = 20$, $\alpha_{95} = 2.7°$, $k = 142$), where D, declination; I, inclination (positive down); N, number of site mean directions; α_{95}, radius of cone of 95% confidence; k, precision parameter of Fisher (1953). Overall, the site mean directions form a tight cluster with a modest "about vertical" to "steep northwesterly" elongation of the population (Fig. 6). The elongation has a crude geographic distribution with the "steep northwesterly" directions (sites 5–10, 14, and 15) coming mostly from the northwestern part of the pluton, the "about vertical" directions (sites 1, 2, 4, 11–13, 17, and 21) coming mostly from the central to southeastern part of the pluton, and the intermediate directions (sites 16, 18–20) coming only from the southeasterly part of the pluton. Noting Ragged Pluton's size of ~5 km in diameter, emplacement depth of 5 to 7 km, and ~300 °C range of blocking temperatures that give collinear direction in magnetite and pyrrhotite, it is clear that the pluton took $>10^4$ years to cool from ~580 to ~280 °C, which assures that secular variation in the Earth's magnetic field has been averaged out (Merrill and McElhinny 1983) in the site mean ChRM directions. Noting, also, that the petrology of Ragged Pluton has not been mapped in detail, the population's elongation most likely reflects the timing and distribution of phases within the pluton. Using a mean site location of 62.0°N, 231.9°E (latitude, longitude), the paleopole for Ragged Pluton is at 73.9°N, 191.4°E ($dp = 5.0°$, $dm = 5.2°$; radii of the oval of 95% confidence along and perpendicular to the site-pole great circle) (Fig. 7).

As an internal check on the dominantly AF-derived mean ChRM direction for Ragged Pluton, the ChRM directions obtained from 19 of the 21 Th step demagnetized specimens in the 260 to 330 °C unblocking temperature range of pyrrhotite were averaged and have a mean direction of $D = 324.3°$, $I = 75.7°$ ($N = 19$, $\alpha_{95} = 5.2°$, $k = 42$), which is not significantly different at 95% confidence from the pluton's ChRM direction (McFadden and Lowes 1981).

Discussion

Ragged Pluton's paleopole of 73.9°N, 191.4°E (A_{95} =

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Fig. 3. Vector component and intensity plots for example specimens on thermal step demagnetization. The first two digits of the sample number indicate the sample site. Solid (open) symbols show the vector in the horizontal north (N) – east (E) – south (S) – west (W) (vertical north (N) – down (D) – south (S) – up (U)) plane with the axes showing the measured intensity (J) as a ratio of the initial NRM (0) intensity (J_0). The intensity drop from ~250 to ~325 °C reflects pyrrhotite and from ~450 to ~585 °C reflects magnetite. The specimen J_0 intensities in mA/m are (*a*) 5.52, (*b*) 6.63, (*c*) 5.22, and (*d*) 4.82.

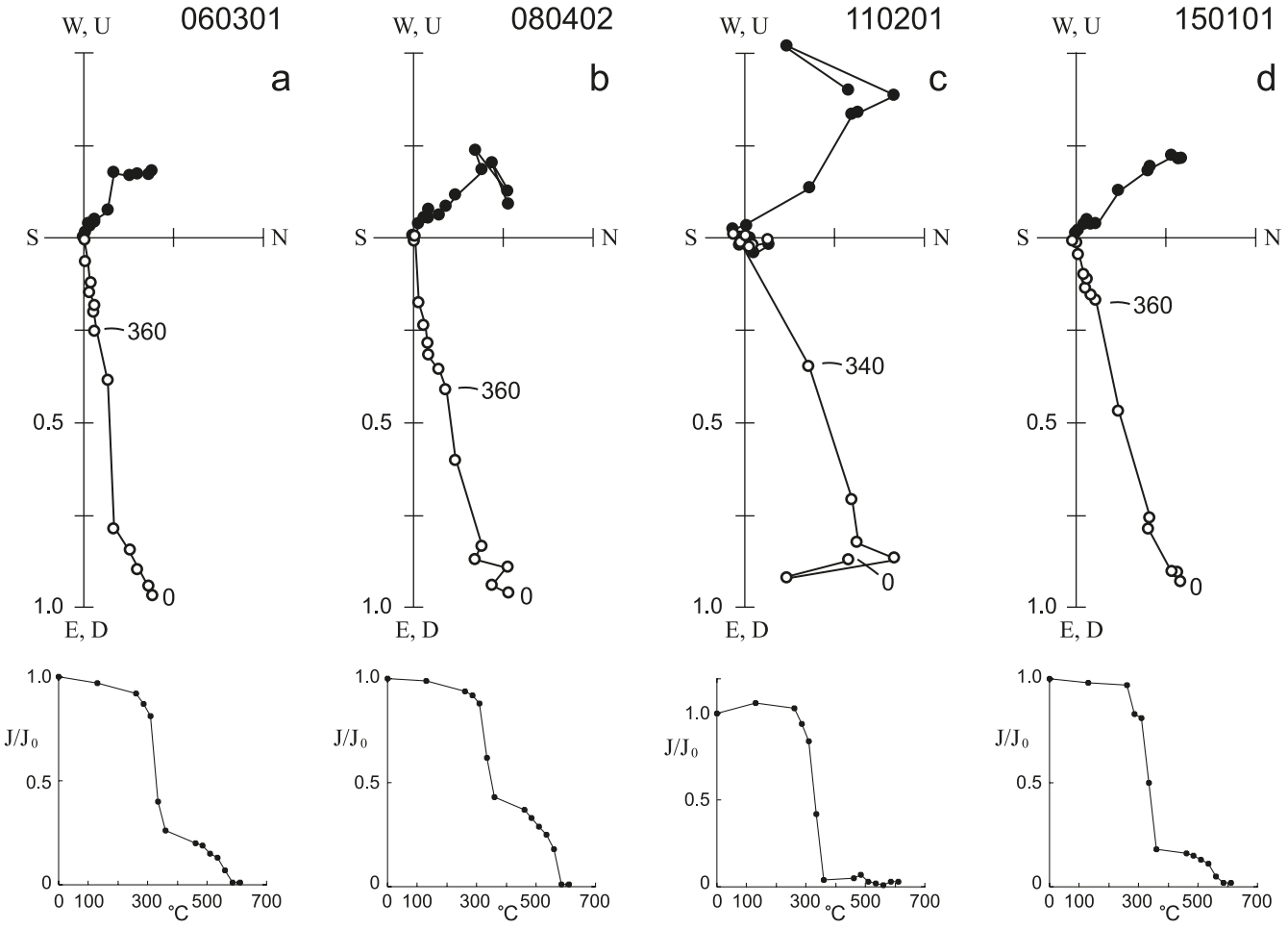
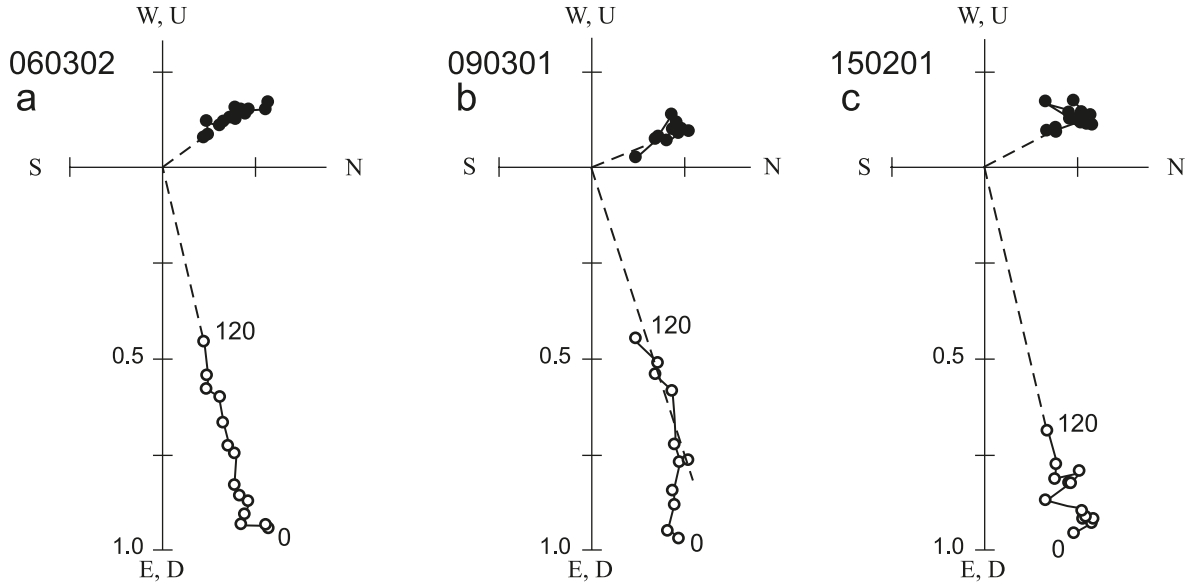


Fig. 4. Vector component plots for example specimens on AF step demagnetization. Conventions as in Fig. 3, except that the initial NRM (0) and 120 mT steps are indicated. The specimen J_0 intensities in mA/m are (*a*) 5.41, (*b*) 2.73, and (*c*) 5.10.



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Fig. 5. Saturation isothermal remanent magnetization (SIRM) results for representative specimens from 10 sites showing (*a*) acquisition of SIRM in a direct field (H_{df}), and (*b*) demagnetization of SIRM in an alternating field (H_{af}) in millitesla (mT). J/J_{max} is the measured remanence intensity (J) as a ratio to the saturation remanence intensity (J_{max}). The dashed lines are at J/J_{max} = 0.9 and at H_{df} = 350 mT.

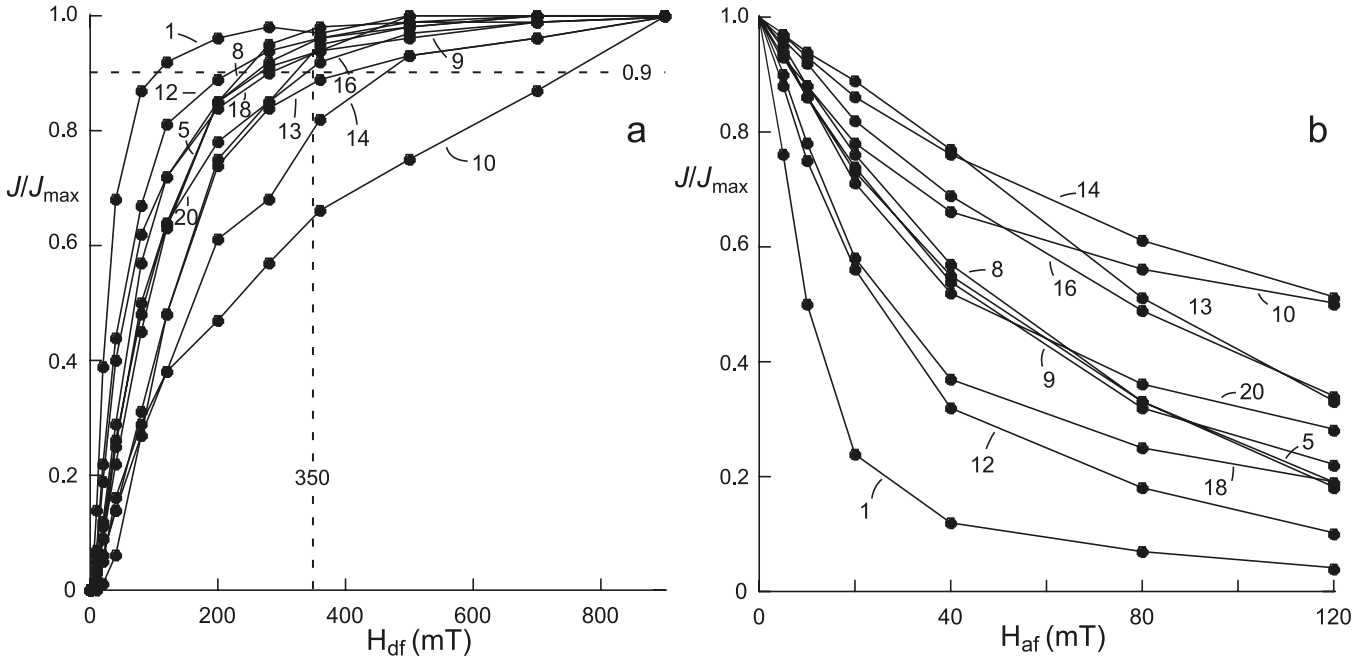


Table 1. Site mean remanence directions.

Site	Specimen end points, remagnetization circles, and omitted from mean			ChRM direction			
				D (°)	I (°)	α_{95} (°)	k
1	6	3	0	290.1	83.2	4.8	116
2	6	3	1	136.3	86.9	5.7	81
3	0	0	10	—	—	—	—
4	3	4	2	291.9	82.0	6.3	93
5	8	0	1	326.9	74.2	3.5	252
6	9	0	0	329.1	72.3	2.3	493
7	10	0	0	331.4	73.0	2.5	385
8	10	0	0	324.8	72.7	1.5	1003
9	10	0	0	331.4	75.9	3.4	205
10	6	3	0	327.2	75.9	3.6	209
11	5	5	2	327.2	85.5	3.9	154
12[a]	6	4	0	322.7	87.2	5.0	94
13[b]	8	2	1	11.5	86.5	5.2	87
14	10	3	0	334.5	74.5	3.0	188
15	8	1	1	315.5	72.8	2.5	412
16	7	4	1	320.0	79.6	4.1	125
17	4	5	0	325.7	87.9	4.5	133
18	6	2	1	327.0	79.4	6.7	69
19	9	3	0	325.7	81.2	3.3	174
20	8	3	0	343.0	77.8	3.1	219
21	4	7	1	258.4	85.8	3.6	160
Mean	20			325.9	80.4	2.7	142

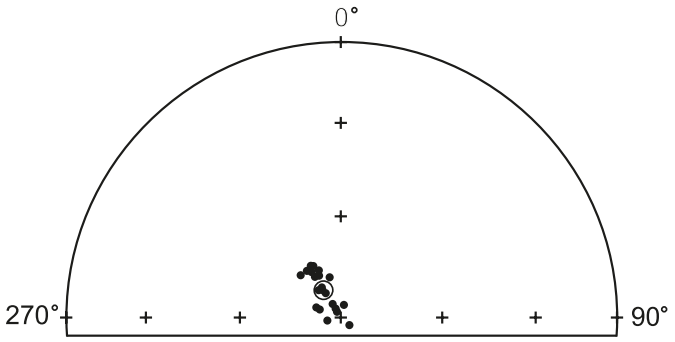
Note: ChRM, characteristic remanent magnetization; D, declination; I, inclination; α_{95}, radius cone of 95% confidence; k, precision parameter of Fisher (1953).

[a]Contact test—monzogranite.

[b]Contact test—host siltstone.

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Fig. 6. Site mean directions on lower hemisphere of an equal-area stereograph with the cone of 95% confidence about the population mean direction.

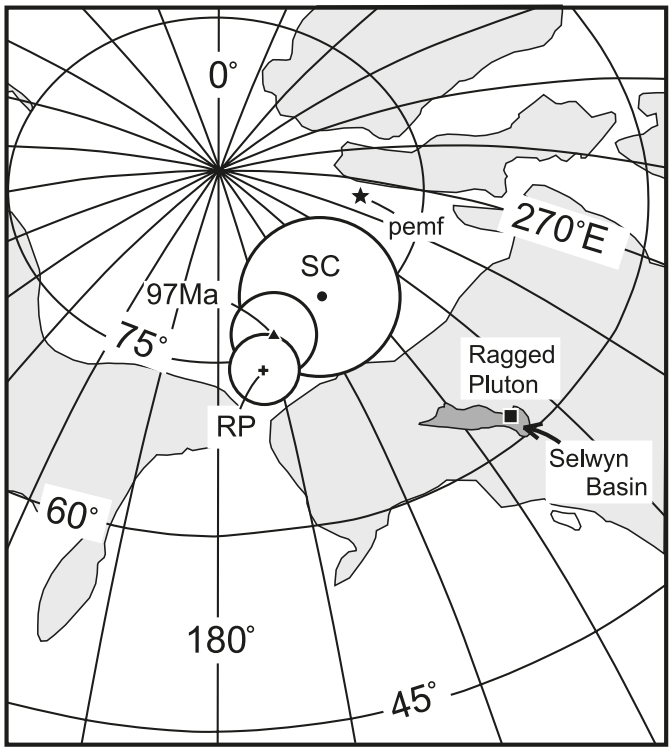


5.1°) is only 2.4° away from the co-eval 97 Ma reference paleopole of 76.3°N, 195.4°E (A_{95} = 6.2°) of Besse and Courtillot (2002) for ancestral North America, showing clearly at >>95% confidence that the two paleopoles are concordant (Fig. 7). The translation and rotation estimates with 95% confidence error estimates (Demarest 1983) are nonsignificant at 0.0° ± 6.4° and 8.1° ± 20.4° counterclockwise, respectively. The relatively large error estimate for rotation is caused by the closeness of the sampling area to the paleopole. This concordancy of Ragged Pluton and North American reference paleopoles shows that the assemblage of rocks forming the eastern Selwyn Basin has not undergone significant translation or rotation with respect to North America since the pluton's emplacement at 97 Ma. This paleomagnetic result provides limits for published tectonic models for the Selwyn Basin.

Paleomagnetic results have been published from the Deadman Pluton in the extreme northwestern corner of the Selwyn Basin (Symons et al. 2006; Fig. 1). This pluton is a massive circular (~180 km^2) 92 ± 1 Ma alkalic felsic stock of the Tombstone Plutonic Suite (Hart et al. 2004) that was emplaced into the metamorphosed Neoproterozoic Hyland Group within the Dawson thrust sheet (Gordey and Makepeace 2003). Deadman Pluton yielded a discordant paleopole that Symons et al. (2006) attributed to ~7° ± 4° of north-side-up tilt of the pluton as the Dawson thrust sheet was pushed northward after 92 Ma up the southward-dipping North Fork and Dawson thrust faults that either intersect or underlie Deadman Pluton at depth.

Also, Irving and Archibald (1990) published a paleopole for the post-deformational mid-Cretaceous Skelly Creek Batholith in southeastern British Columbia that intrudes deformed strata of the North American continental margin (Fig. 1). The batholith is composed of massive hornblende–biotite granodiorite, is exposed over ~280 km^2, intrudes metasedimentary strata of the Purcell anticlinorium and is dated by K–Ar biotite at 94 to 99 Ma. Contact metamorphic assemblages suggest the batholith has not been tilted by >5° in any direction. Paleomagnetic sampling at 17 sites, mainly from dioritic and metasedimentary xenoliths and from the pluton's contact metamorphic aureole, gave a ChRM direction of D = 349°, I = 74° (N = 13, α_{95} = 6°, k = 53). Using a mean site location of 49.4°N, 243.4°E, the batholith's paleopole is located at 78°N, 216°E (A_{95} = 10°) (Fig. 7) without a tectonic correction, which is not statistically different

Fig. 7. Locations of the Ragged Pluton (RP), 97 Ma North American reference (Besse and Courtillot 2002) and the Skelly Creek (SC) paleopoles with their 1σ cones of confidence. Intersection of the 1σ cones of confidence shows a close approximation that the paleopoles are not significantly different at 95% (2σ) confidence. The star symbol is present Earth's magnetic field (pemf) direction.



at 95% confidence also from the 97 Ma North American reference pole (Besse and Courtillot 2002; Fig. 7). The concordancy of the Ragged Pluton, Skelly Creek Batholith, and North American reference paleopoles supports Irving and Archibald's (1990) conclusion that the Skelly Creek region in southeastern British Columbia "was close to the margin of ancestral North America in mid-Cretaceous time, essentially where it is today" and extends their conclusion northwards to include the Selwyn Basin.

There have been a series of paleomagnetic studies on the 70 Ma Carmacks Group flood volcanics in the Yukon. Marquis and Globerman (1988) reported results from 18 flows and concluded that the IMB terranes had been translated by 1500 ± 950 km northward relative to North America. Butler (1990) contended that an insufficient number of time-independent flows had been sampled to properly average out secular variation, giving an anomalous result. Marquis et al. (1990) replied, pointing out that Butler's (Butler 1990) re-analysis of their data gave a displacement that was only about 20% smaller. Subsequently five of the 18 flows were dated radiometrically and shown to be too old to be part of the Carmacks Group (Mihalynuk et al. 1992). Next, Johnston et al. (1996; see also Wynne et al. 1998) measured an additional 13 sites, giving 26 in total, and concluded that the IMB had been translated northward by 1900 ± 700 km between 70 and 53 Ma relative to ancestral North America, thereby eliciting concern over the anomalously large magnitude of the estimate (Harris et al.

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1999; Symons et al. 2005). Recently Enkin et al. (2006) extended the Carmacks paleomagnetic study by reporting acceptable data from an additional 15 of 22 tested sites at Solitary Mountain. Using the data from all 41 Carmacks volcanics sites, Enkin et al. (2006) concluded that the IMB and Y–T terranes had been translated northward relative to North America by 1950 ± 600 km after 68 Ma, of which ~425 km could be accommodated by dextral motion on the Tintina Fault (Gabrielse et al. 2006), requiring the remaining ~1525 km to be accommodated by an unidentified structure(s) to the east of the Selwyn Basin. Gladwin and Johnston (2006) pointed out that the contact aureole of the ~105 Ma Glenlyon Batholith in the Cassiar Terrane extends across the Tummel Fault Zone into the Y–T Terrane (Fig. 1), thus pinning the Y–T Terrane to the Cassiar Terrane before extrusion of the Carmacks Group. Further, Gladwin and Johnston (2006) noted that the Cassiar Terrane has traditionally been considered to be an integral part of the ancient North American miogeocline (Gordey and Anderson 1993) and that coeval plutons related to the Glenlyon Batholith also pin the Cassiar Terrane to the Selwyn Basin at ~105 Ma. These related intrusions include a 103 ± 2 Ma leucogabbro in the Tummel Fault Zone, a 105 ± 4 Ma hornfels from the Truit Creek map area that includes part of the Glenlyon Batholith, the 109 ± 3 Ma d'Abbadie pluton about 15 km to the southwest, and a 108 Ma porphyry at Clear Lake about 70 km to the northwest. Thus Gladwin and Johnston (2006) suggested that either the paleomagnetic results from the Carmacks volcanics are unreliable for some as-yet-unexplained reason or that both the Cassiar Terrane and Selwyn Basin have been translated northward (poleward) also by 1525 ± 600 km, excluding Tintina fault displacement, relative to ancestral North America on some as-yet-unidentified fault structure to the east of the Selwyn Basin. The finding of a concordant paleopole for the 97 Ma Ragged Pluton supports the notion that the eastern Selwyn Basin was not translated northward after either 97 or ~68 Ma. However, the Ragged Pluton paleopole does not preclude in itself the possibility that some as-yet-unidentified major fault exists that cuts across the Selwyn Basin between the pluton and the Glenlyon Batholith area.

Bardoux and Irving (1989) and Symons and Wellings (1989) (see also Symons et al. 2000) were first to point out from paleomagnetic evidence that the IMB terranes, as part of Baja British Columbia, had accreted to stable North America by ~52 Ma and were no longer being translated northward. There are now Eocene results available from seven IMB paleomagnetic collections and one Y–T collection throughout British Columbia and Yukon (McCausland et al. 2006) that come from 240 sites in volcanics, plutons, and dikes with a mean age of 51.3 ± 2.9 Ma. The eight collections yield a nonsignificant mean northward (poleward) post-Early Eocene translation of 1.2° ± 4.8° or 130 ± 530 km that is consistent with the 400–430 km estimate of Gabrielse et al. (2006) for dextral Paleocene–Eocene (<67 Ma) displacement on the Tintina Fault (Fig. 1). Using the 51 Ma age, northward translation of the Carmacks volcanics by 1950 ± 600 km after 68 Ma as proposed by Enkin et al. (2006) would have had to be accomplished in ~17 Ma, requiring a minimum (paleolatitudinal) translation rate of 11.5 ± 3.5 cm/a for the IMB and Y–T terranes. Terrane translation rates as high as ~18 cm/a are known in the paleomagnetic and seafloor magnetic record (Meert et al. 1993). However, the known velocity for the Kula Plate, which is the only probable transport mechanism for the Carmacks volcanics, is ~13 cm/a for the 68 to 51 Ma interval (Engebretson et al. 1985; Kelley 1993). Thus the translation rate for the Carmacks volcanics requires at least a 90% coupling—a very high degree of coupling—between the IMB and Y–T terranes and the Kula Plate prior to collision.

Paleomagnetic evidence for post-mid-Cretaceous tectonism in the northern Cordillera to the east of Ragged Pluton is minimal. Symons et al. (2003) reported paleomagnetic results from three 42 ± 1 Ma post-tectonic plutons of the Beaver River alkalic complex that intrude strata of the Mackenzie Platform of southeasternmost Yukon (Fig. 1). These plutons gave a well-defined paleopole that was concordant with the co-eval North American reference paleopole, thereby showing that the Foreland Belt of the Mackenzie Platform has not been rotated or translated significantly since the mid-Eocene. The Beaver River result, however, does not provide tectonic evidence for the more critical mid-Cretaceous to mid-Eocene time interval.

Symons et al. (2003) observed that Oligocene–Pliocene and Eocene paleomagnetic collections in the accreted IMB and eastern Coast Mountain terranes recorded clockwise ChRM rotations of 13° ± 5° and 24° ± 10°, respectively, relative to stable North America. They attributed these rotations to the allochthonous terranes being obducted and driven progressively up and over a lower crustal ramp of pericratonic and (or) cratonic North American terranes by the subducting Farallon plate since the mid-Eocene. Subsequently Symons et al. (2004, 2005) extended the tectonic motion study in a review of all Mesozoic and Cenozoic paleomagnetic data for the IMB and Y–T terranes. From the results of 15 paleomagnetic studies in Middle Eocene and younger rock units, they found that these terranes were rotated clockwise at a steady rate of 0.29° ± 0.11° per million years or by 16° ± 6° since 54 Ma. Symons et al. (2005) cited evidence from seismic (reflection, refraction and earthquake), heat flow, gravity, global positioning system, balanced geological cross-section, and geochemical studies in support of the paleomagnetic estimate of 305 ± 125 km for the ENE–WSW "shortening" across the Selwyn Basin and Mackenzie Mountains in the southern Yukon between the IMB and ancestral North America. This "shortening" was attributed to thin-skin tectonics as the IMB terranes moved east-northeastward over the underlying pericratonic Y–T Terrane and other autochthonous terranes of ancestral North America. The concordant paleopole determined for Ragged Pluton is permissive for this thin-skin model as the pluton was emplaced into the overridden Selwyn Basin. The concordant Ragged Pluton paleopole does, however, preclude a thick-skin tectonic model as it would require a significant clockwise rotation of the ChRM declination in the pluton. Since the additional 35° ± 14° of clockwise rotation and 8.3° ± 7.0° (915 ± 775 km) of northward translation that is observed in the IMB terranes between ~102 and 54 Ma is not observed in the Ragged Pluton's paleopole, it is evident that the eastern Selwyn Basin did not travel as part of Baja British Columbia during this time interval as suggested by Enkin et al. (2006). Rather, Ragged Pluton's paleopole is

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consistent with the hypothesis proposed by Symons et al. (2005) and McCausland et al. (2006) that the IMB terranes were travelling with Baja British Columbia on the Kula Plate, whereas Ragged Pluton was traveling with the North American Plate. Finally, comparison of the six paleopoles available from plutons in the Y–T Terrane (two Early Jurassic, one mid-Cretaceous, two Late Cretaceous, and one Paleocene pluton) do not show consistent or significant paleolatitudinal displacements with respect to ancestral North America, excluding dextral displacement of ~425 km on the Tintina fault, which suggests that the Y–T Terrane was attached to North America since the Early Jurassic in about its present location, and was not displaced ~1950 km northward during the Paleocene–Eocene as Enkin et al. (2006) hypothesized. This observation led Symons et al. (2005) and McCausland et al. (2006) to deem the Y–T Terrane to be para-autochthonous, an interpretation that is consistent with the determination of a concordant paleopole for Ragged Pluton.

Conclusions

- The 97 Ma Ragged Pluton of the Tungsten Plutonic Suite in the eastern Selwyn Basin carries a stable ChRM that resides in pseudosingle to single domain pyrrhotite and magnetite that yield a colinear remanence direction. Thus the ChRM is deemed to be a primary thermoremanent magnetization that was acquired on cooling of the pluton.
- The paleopole derived for Ragged Pluton at 73.9°N, 191.4°E (A_{95} = 5.1°) is ~2.4° away from the co-eval 97 Ma reference paleopole for ancestral North America, showing that the two paleopoles are concordant with >>95% confidence. Their concordancy shows that the eastern Selwyn Basin has not undergone significant rotation or translation since 97 Ma.
- The concordant paleopole for Ragged Pluton conflicts with tectonic models that advocate that the Selwyn Basin was part of the far-traveled Baja British Columbia microplate that moved ~2000 km northward during the Paleocene–Eocene relative to ancestral North America.

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References

Bailey, R.C., and Halls, H.C. 1984. Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data. Journal of Geophysics, **54**: 174–182.

Bardoux, M., and Irving, E. 1989. Paleomagnetism of Eocene rocks of the Kelowna and Castlegar areas, British Columbia: studies in determining paleohorizontal. Canadian Journal of Earth Sciences, **26**: 829–844.

Besse, J., and Courtillot, V. 2002. Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr. Journal of Geophysical Research, **107**(B11): 2300–2331. doi:10.1029/2000JB000050.

Butler, R.F. 1990. Northward motion of the Whitehorse Trough: paleomagnetic evidence from the Upper Cretaceous Carmacks Group. Canadian Journal of Earth Sciences, **27**: 614–616. [Discussion.]

Demarest, H.H. 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data. Journal of Geophysical Research, **88**: 4321–4328. doi:10.1029/JB088iB05p04321.

Engebretson, D.C., Cox, A., and Gordon, R.G. 1985. Relative motions between oceanic and continental plates in the Pacific Basin. Geological Society of America, Special Paper 206, pp. 1–59.

Enkin, R.J., Johnston, S.T., Larson, K.P., and Baker, J. 2006. Paleomagnetism of the 70 Ma Carmacks Group at Solitary Mountain, Yukon, confirms and extends controversial results: Further evidence for the Baja British Columbia model. *In* Paleogeography of the North American Cordillera: evidence for and against large-scale displacements. *Edited by* J.W. Haggart, R.J. Enkin, and J.W.H. Monger. Geological Association of Canada, pp. 221–232.

Fisher, R.A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, Series A: Mathematical and Physical Sciences, **217**: 295–305. doi:10.1098/rspa.1953.0064.

Gabrielse, H. 1967. Tectonic evolution of the northern Canadian Cordillera. Canadian Journal of Earth Sciences, **4**: 271–298.

Gabrielse, H., Monger, J.W.H., Wheeler, J.O., and Yorath, C.J. 1991. Morphogeological belts, tectonic assemblages and terranes. *In* Geology of the Cordilleran Orogen in Canada. *Edited by* H. Gabrielse and C.J. Yorath. Geological Survey of Canada, Geology of Canada. no. 4, Part A, pp. 15–28.

Gabrielse, H., Murphy, D.C., and Mortensen, J.K. 2006. Cretaceous and Cenozoic dextral orogen-parallel displacements, magmatism, and paleogeography, north-central Canadian Cordillera. *In* Paleogeography of the North American Cordillera: evidence for and against large-scale displacements. *Edited by* J.W. Haggart, R.J. Enkin, and J.W.H. Monger. Geological Association of Canada. pp. 255–276.

Gladwin, K., and Johnston, S.T. 2006. Mid-Cretaceous pinning of accreted terranes to miogeoclinal assemblages in the northern Cordillera: Irreconcilable with paleomagnetic data? *In* Paleogeography of the North American Cordillera: evidence for and against large-scale displacements. *Edited by* J.W. Haggart, R.J. Enkin, and J.W.H. Monger. Geological Association of Canada. pp. 299–306.

Gordey, S.P., and Anderson, R.G. 1993. Evolution of the northern Cordilleran miogeocline, Nahanni map area (105I), Yukon and Northwest Territories. Geological Survey of Canada, Ottawa, Ont., Memoir 428, 214 p.

Gordey, S.P., and Makepeace, A.J. (*Editors*). 2003. Yukon Digital Geology (version 2). Yukon Geological Survey, Open File 2003-9(D) (also known as Geological Survey of Canada, Open File 1749). [2 CD-ROMs.]

Harris, M.J., Symons, D.T.A., Blackburn, W.H., and Hart, C.J.R. 1999. Paleomagnetic and geobarometric study of the Late Cretaceous Mount Lorne Stock, southern Yukon Territory. Canadian Journal of Earth Sciences, **36**: 905–915. doi:10.1139/cjes-36-6-905.

Hart, C.J.R., Goldfarb, R.J., Lewis, L.L., and Mair, J.L. 2004. The northern Cordilleran mid-Cretaceous plutonic province: Ilmenite/magnetite-series granitoids and intrusion-related mineralisation. Resource Geology, **54**(3): 253–280.

Hart, C.J.R., Mair, J.L., Goldfarb, R.J., and Groves, D.I. 2005.

© 2008 NRC Canada

Source and redox controls of intrusion-related metallogeny, Tombstone–Tungsten Belt, Yukon, Canada. *In* 5th Hutton Symposium Volume on Origin of granites and related rocks. *Edited by* S. Ishiahara, W.E. Stephens, S.L. Harley, M. Arima, and T. Nakajima. Geological Society of America, Special Paper 389, Royal Society of Edinburgh, UK. pp. 339–356.

Hart, C.J.R., and Lewis, L.L. 2006. Gold mineralization in the upper Hyland River area: A non-magmatic origin. *In* Yukon exploration and geology 2005. *Edited by* D.S. Emond, G.D. Bradshaw, L.L. Lewis, and L.H. Weston. Yukon Geological Survey, pp. 109–125.

Heffernan, R.S. 2004. Temporal, geochemical, isotopic, and metallogenic studies of mid-Cretaceous magmatism in the Tintina Gold Province, southeastern Yukon and southwestern Northwest Territories, Canada. Unpublished M.Sc. thesis, The University of British Columbia, Vancouver, B.C., 85 p.

Irving, E., and Archibald, D.A. 1990. Bathozonal tilt corrections to paleomagnetic data from mid-Cretaceous plutonic rocks: examples from the Omineca Belt, British Columbia. Journal of Geophysical Research, **95**: 4579–4585. doi:10.1029/JB095iB04p04579.

Irving, E., Wynne, P.J., Thorkelson, D.J., and Schiarizza, P. 1996. Large (1000 to 4000 km) northward movements of tectonic domains in the northern Cordillera, 83 to 45 Ma. Journal of Geophysical Research, **101**: 17901–17916. doi:10.1029/96JB01181.

Johnston, S.T., Wynne, P.J., Francis, D., Hart, C.J.R., Enkin, R.J., and Engebretson, D.C. 1996. Yellowstone in Yukon: the Late Cretaceous Carmacks Group. Geology, **24**: 997–1000. doi:10.1130/0091-7613(1996)024<0997:YIYTLC>2.3.CO;2.

Kelley, K. 1993. Relative motions between North America and oceanic plates of the Pacific basin during the past 130 million years. M.Sc. thesis, Western Washington University, Bellingham, Wash.

Kirschvink, J.L. 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

Mair, J.L., Hart, C.J.R., and Stephens, J.R. 2006. Deformation history of the western Selwyn Basin, Yukon, Canada: Implication for orogen evolution and mid-Cretaceous magmatism. Geological Society of America Bulletin, **118**: 304–323. doi:10.1130/B25763.1.

Marquis, G., and Globerman, B.R. 1988. Northward motion of the Whitehorse Trough: paleomagnetic evidence from the Upper Cretaceous Carmacks Group. Canadian Journal of Earth Sciences, **25**: 2005–2016.

Marquis, G., Irving, E., and Globerman, B.R. 1990. Northward motion of the Whitehorse Trough: paleomagnetic evidence from the Upper Cretaceous Carmacks Group. Canadian Journal of Earth Sciences, **27**: 617–618. [Reply.]

McCausland, P.J.A., Symons, D.T.A., Hart, C.J.R., and Blackburn, W.H. 2006. Assembly of the northern Cordillera: new paleomagnetic evidence for coherent, moderate Jurassic to Eocene motion of the Intermontane Belt and Yukon-Tanana terranes. *In* Paleogeography of the North American Cordillera: evidence for and against large-scale displacements. *Edited by* J.W. Haggart, R.J. Enkin, and J.W.H. Monger. Geological Association of Canada, pp 147–170.

McFadden, P.L., and Lowes, F.J. 1981. The discrimination of mean directions drawn from Fisher distributions. Geophysical Journal of the Royal Astronomical Society, **67**: 19–33.

Meert, J.G., Van der Voo, R., Powell, C.M., Li, Z.-X., McElhinny, M.W., Chen, Z., and Symons, D.T.A. 1993. A plate-tectonic speed limit? Nature, **363**: 216–217. doi:10.1038/363216a0.

Merrill, R.T., and McElhinny, M.W. 1983. The Earth's Magnetic Field. Academic Press, London, UK., 401 p.

Mihalynuk, M.G., Smith, M.T., Gabites, J.E., Runkle, D., and Lefebure, D. 1992. Age of emplacement and basement character of the Cache Creek Terrane as constrained by new isotopic and geochemical data. Canadian Journal of Earth Sciences, **29**: 2463–2477.

Symons, D.T.A., and Wellings, M.R. 1989. Paleomagnetism of the Eocene Kamloops Group and the cratonization of Terrane I of the Canadian Cordillera. Canadian Journal of Earth Sciences, **26**: 821–828.

Symons, D.T.A., Harris, M.J., Gabites, J.E., and Hart, C.J.R. 2000. Eocene (51 Ma) end to northward translation of the Coast Plutonic Complex: Paleomagnetism and K–Ar dating of the White Pass dikes. Tectonophysics, **326**: 93–109. doi:10.1016/S0040-1951(00)00148-7.

Symons, D.T.A., Erdmer, P., and McCausland, P.J.A. 2003. New 42 Ma cratonic North American paleomagnetic pole from the Yukon underscores another Cordilleran paleomagnetism–geology conundrum. Canadian Journal of Earth Sciences, **40**: 1321–1334. doi:10.1139/e03-047.

Symons, D.T.A., Hart, C.J.R., Harris, M.J., and McCausland, P.J.A. 2004. Paleomagnetism of the Eocene Flat Creek Pluton, Yukon: tectonics of the Intermontane Terranes and Mackenzie Mountains *In* Orogenic curvature: integrating paleomagnetic and structural analyses. *Edited by* A.J. Sussman and A.B. Weil. Geological Society of America, Special Paper 383, pp. 187–204.

Symons, D.T.A., Harris, M.J., and McCausland, P.J.A. 2005. Mesozoic-Cenozoic paleomagnetism of the Intermontane and Yukon–Tanana terranes, Canadian Cordillera. Canadian Journal of Earth Sciences, **42**: 1163–1185. doi:10.1139/e04-086.

Symons, D.T.A., Harris, M.J., Hart, C.J.R., and McCausland, P.J.A. 2006. Paleomagnetism of the ~91 Ma Deadman pluton: Post-mid-Cretaceous tectonic motion in central Yukon. *In* Yukon exploration and geology 2005. *Edited by* D.S. Emond, G.D. Bradshaw, L.L. Lewis, and L.H. Weston. Yukon Geological Survey, pp. 299–313.

Wynne, P.J., Enkin, R.J., Baker, J., Johnston, S.T., and Hart, C.J.R. 1998. The big flush: paleomagnetic signature of a 70 Ma regional hydrothermal event in displaced rocks of the northern Canadian Cordillera. Canadian Journal of Earth Sciences, **35**: 657–671. doi:10.1139/cjes-35-6-657.

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